

CS301 Project Description

This project has three stages. You may focus on **regression**, **classification**, or both, covering the full ML pipeline from preprocessing to deployment.

Stage 1: Data Preprocessing & Initial Modeling

Dataset Requirements: Select a dataset with at least 12 features, including a minimum of two categorical and two numerical features. Define the target variable as either numerical (regression) or categorical (classification).

Preprocessing: Clean data (missing values, duplicates, errors) and perform EDA using visualizations to understand feature–target relationships.

Initial Modeling:

- **Regression:** Select features via correlation, train multiple regression, evaluate using **R²** and **RMSE**.
- **Classification:** Train a classifier (e.g., Logistic Regression), evaluate using **Accuracy** and **F1-score**.

Stage 2: Model Optimization & Comparative Analysis

Advanced Modeling: Implement **two additional models**.

Hyperparameter Tuning (if applicable): Use **Grid/Randomized Search + k-fold cross-validation**.

Feature Engineering: Apply feature selection, polynomial feature transformation, or regularization(optional).

Performance Comparison: Compare models using the two evaluation metrics with tables/visuals; identify the **best model**.

Stage 3: Application Development & Deployment [\[Sample Link\]](#)

Data Upload: Upload CSV with automated preprocessing.

Dynamic Target Selection: Dropdown for target; show **average values (regression)** or **class distribution (classification)**.

Feature Selection & Visualization: Select features and display correlation bar chart.

Training Pipeline: Use a **scikit-learn Pipeline** (missing values + one-hot encoding) to train the best model.

Prediction Interface: Enable manual input for real-time predictions.

Deployment: Host on **AWS, GCP, or Heroku**.

Submission Requirements

- **Code:** Clean, well-documented codes in a single zip file with all required files.
- **Report (maximum 3 pages):**
 - Problem definition and objective
 - Dataset description and preprocessing steps
 - Exploratory Data Analysis (key insights with visuals)
 - Model selection and justification
 - Evaluation metrics and performance comparison
 - Final model choice with reasoning (strengths, limitations, and trade-offs)
 - URL of the deployed application.
- **Video Demonstration:** Record a brief video demonstrating the application interface and its functionality.

Upload the code, report, and video to a Google Drive folder and submit the folder URL. Ensure that access permissions allow anyone with the link to view the contents.